



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to NANDHINI AGENCY

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

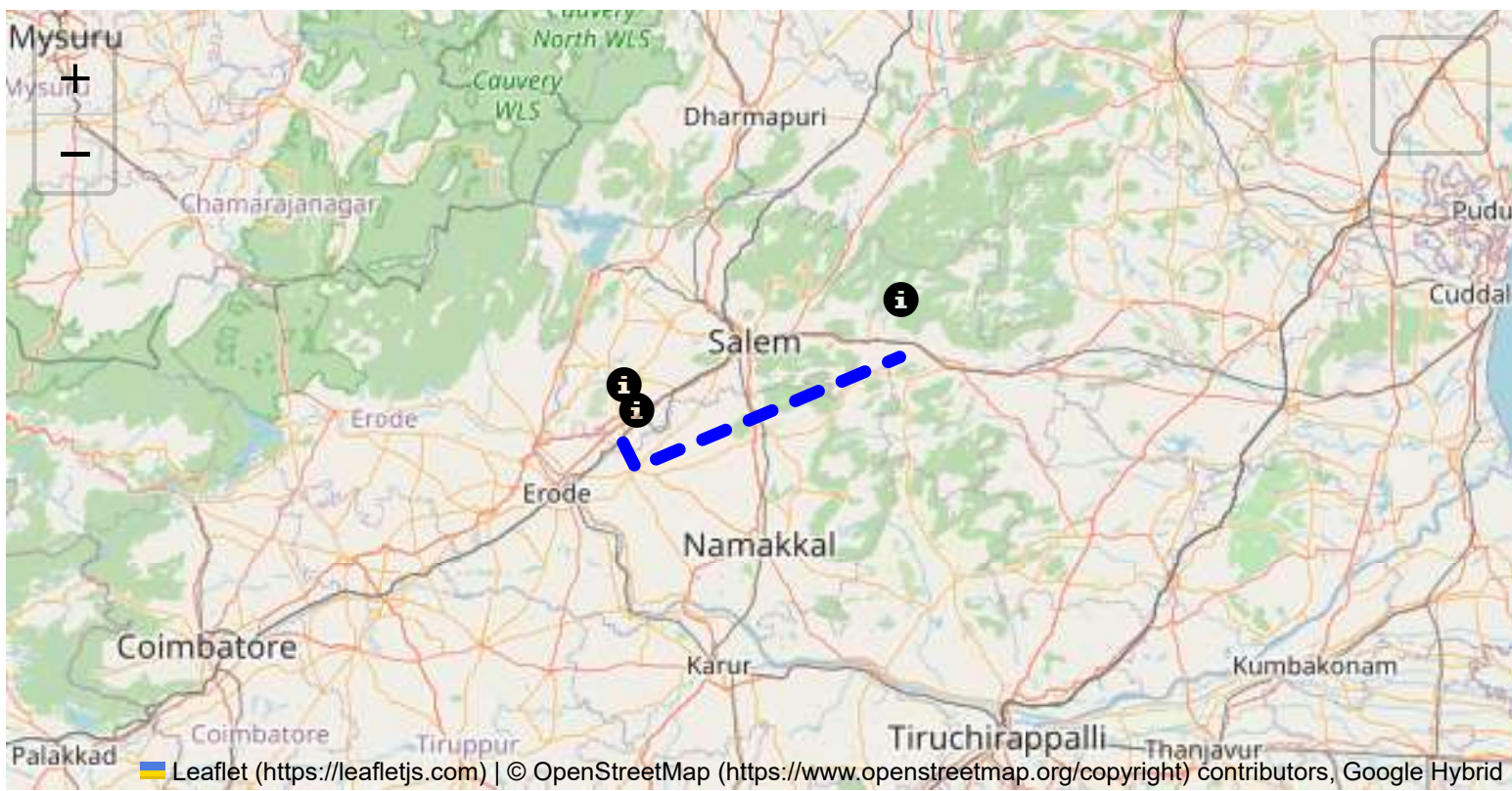
## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

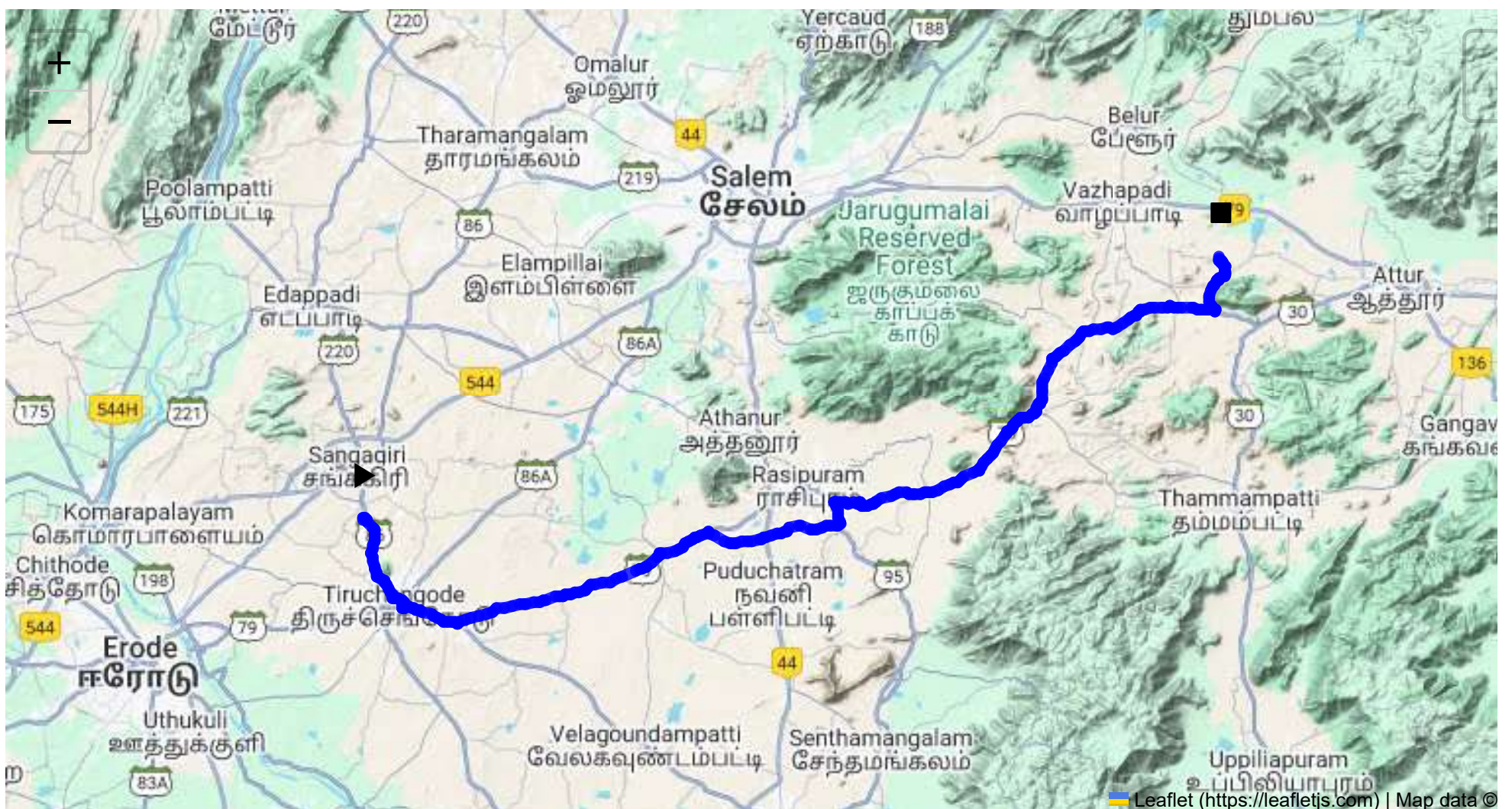
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 88.15 km**  
**Estimated Duration: 1.9 hours**  
**Adjusted Duration (Heavy Vehicle): 2.4 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.61912, 78.472854)**



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

The route from Sangagiri to Muthakavandanur via Tiruchengode - Salem Main Road is approximately 88.15 kilometers. This route passes through several towns and semi-urban areas, including Santhapattai and Thokkavadi. The journey primarily involves traveling on state highways with varying traffic conditions. Key intersections, bypasses, and direct highway stretches are part of this route.



## 2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Patterns:** Tamil Nadu generally experiences a tropical climate, with peak temperatures between April and June. The monsoon season spans from October to December.
- **Hazards:** During monsoon months, there is a risk of heavy rains leading to slippery roads and reduced visibility. Flooding in low-lying areas and waterlogging can occur, though less frequently along raised highway sections.

## 3. Analysis of Traffic Patterns

- **Peak Hours:** Typically, the route experiences heavy traffic during morning (8:00 AM - 10:00 AM) and evening (5:00 PM - 7:00 PM) rush hours.
- **Congestion-Prone Areas:** Town centers like Tiruchengode and Salem are prone to congestion. The exits and entrances of towns may have bottlenecks due to narrow road sections and intersections.

## 4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Roads are generally in moderate to good condition, but some stretches may have potholes or uneven patches, especially during or after the rainy season.
- **Infrastructure:** This route has limited roadside facilities for trucks. Bridges and culverts are structurally sound but occasional maintenance work may cause temporary delays.

## 5. Suggestions for Alternative Routes for Emergencies

- In case of major roadblocks, the driver could consider rerouting through local roads between Salem and Namakkal, providing a detour option to bypass congestion or accidents.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

- Transport of hazardous materials is regulated with restrictions on certain timings within town limits, often requiring early morning or late-night travel.
- Specific permits and documentation are necessary, and drivers are expected to comply with safety protocols like proper labeling and safety equipment.

## 7. Overview of Historical Incidents

- **Past Incidents:** There have been occasional incidents involving heavy vehicles, often due to road quality issues, especially during rains or due to driver fatigue leading to rollovers or collisions.
- **Emergency Preparedness:** Emergency services are often strained; ensuring regular rest stops for driver alertness is recommended.

## 8. Environmental Considerations and Sensitive Areas

- The route does not run through major ecological reserves, but there are agricultural lands and water bodies nearby. Spills or accidents involving hazardous materials could impact these areas.

## 9. Analysis of Communication Coverage

- **Coverage:** Mobile network coverage is generally good across the route, but certain remote sections might experience brief coverage dropout, especially in less populated areas.

## 10. Estimated Emergency Response Times

- **Urban Areas:** Approximately 15-30 minutes response time in towns like Salem or Tiruchengode, due to proximity to emergency services.
- **Rural Sections:** Response might take 45 minutes to an hour, as facilities are farther apart in less dense areas.

## 12. Overall Summary of Risk Assessment

- **Risk Level:** Moderate, due to road conditions and traffic congestion. Weather conditions can exacerbate risks, particularly during the monsoon.
- **Mitigation Strategies:** Drivers should be properly trained in hazardous materials handling and emergency procedures. Awareness of local traffic laws and regular vehicle maintenance are essential. Flexible scheduling to avoid peak traffic hours and ensuring proper rest stops for drivers can help manage fatigue-related risks.

Overall, while this route provides a viable connection between Sangagiri and Muthakavandanur, careful attention to weather updates, traffic conditions, and emergency protocols is essential to minimize any potential hazards.

### Risk Assessment - Turns

|   | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|---|------------|------------|--------------------|-------------|
| 0 | Roundabout | High       | 11.38395, 77.89480 | 15 KM/Hr    |
| 1 | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 2 | Turn       | High       | 11.44019, 77.87547 | 15 KM/Hr    |
| 3 | Turn       | Medium     | 11.39685, 77.88861 | 30 KM/Hr    |
| 4 | Turn       | Medium     | 11.38346, 77.89975 | 30 KM/Hr    |
| 5 | Turn       | Medium     | 11.38302, 77.89995 | 30 KM/Hr    |
| 6 | Turn       | High       | 11.38188, 77.90257 | 15 KM/Hr    |
| 7 | Blind Spot | Blind Spot | 11.37841, 77.90140 | 10 KM/Hr    |
| 8 | Turn       | Medium     | 11.36745, 77.93951 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 9  | Turn       | Medium     | 11.41361, 78.08073 | 30 KM/Hr    |
| 10 | Turn       | High       | 11.41453, 78.08062 | 15 KM/Hr    |
| 11 | Turn       | Medium     | 11.41476, 78.08094 | 30 KM/Hr    |
| 12 | Turn       | High       | 11.42956, 78.11590 | 15 KM/Hr    |
| 13 | Turn       | Medium     | 11.42185, 78.13013 | 30 KM/Hr    |
| 14 | Turn       | Medium     | 11.42450, 78.15278 | 30 KM/Hr    |
| 15 | Turn       | High       | 11.43542, 78.20754 | 15 KM/Hr    |
| 16 | Turn       | Medium     | 11.45075, 78.20391 | 30 KM/Hr    |
| 17 | Turn       | High       | 11.45099, 78.20397 | 15 KM/Hr    |
| 18 | Turn       | High       | 11.44946, 78.22270 | 15 KM/Hr    |
| 19 | Turn       | High       | 11.44793, 78.22320 | 15 KM/Hr    |
| 20 | Turn       | Medium     | 11.44753, 78.22443 | 30 KM/Hr    |
| 21 | Turn       | Medium     | 11.45361, 78.23597 | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.49628, 78.32342 | 30 KM/Hr    |
| 23 | Turn       | Medium     | 11.50514, 78.33092 | 30 KM/Hr    |
| 24 | Turn       | Medium     | 11.50868, 78.33972 | 30 KM/Hr    |
| 25 | Turn       | Medium     | 11.51207, 78.34431 | 30 KM/Hr    |
| 26 | Turn       | Medium     | 11.51872, 78.34817 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.56722, 78.37819 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.56945, 78.39881 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 11.58331, 78.44344 | 30 KM/Hr    |
| 30 | Blind Spot | Blind Spot | 11.58139, 78.46967 | 10 KM/Hr    |
| 31 | Turn       | High       | 11.60301, 78.47172 | 15 KM/Hr    |
| 32 | Turn       | High       | 11.60298, 78.47242 | 15 KM/Hr    |
| 33 | Turn       | Medium     | 11.60690, 78.47738 | 30 KM/Hr    |
| 34 | Turn       | Medium     | 11.60757, 78.47727 | 30 KM/Hr    |
| 35 | Turn       | High       | 11.60801, 78.47683 | 15 KM/Hr    |
| 36 | Turn       | High       | 11.61147, 78.47723 | 15 KM/Hr    |
| 37 | Turn       | High       | 11.61194, 78.47606 | 15 KM/Hr    |
| 38 | Turn       | High       | 11.61263, 78.47588 | 15 KM/Hr    |
| 39 | Turn       | High       | 11.61278, 78.47512 | 15 KM/Hr    |
| 40 | Turn       | High       | 11.61424, 78.47467 | 15 KM/Hr    |
| 41 | Turn       | Medium     | 11.61468, 78.47381 | 30 KM/Hr    |
| 42 | Turn       | Medium     | 11.61499, 78.47361 | 30 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 43 | Turn      | High       | 11.61565, 78.47373 | 15 KM/Hr    |
| 44 | Turn      | High       | 11.61597, 78.47302 | 15 KM/Hr    |

### Emergency Locations

|    | type     | name                                | coordinates            | speed_limit | risk_level |
|----|----------|-------------------------------------|------------------------|-------------|------------|
| 0  | hospital | Tiruchengode, Goverment Hospital    | 11.3903328, 77.8920627 | 30 km/h     | Medium     |
| 1  | hospital | SPM Medical Centre,Tiruchengode     | 11.3881331, 77.8931963 | 30 km/h     | Medium     |
| 5  | hospital | Jayaa Hospital                      | 11.3848223, 77.9021886 | 30 km/h     | Medium     |
| 10 | hospital | Government Hospital, Namagiripettai | 11.4569484, 78.265263  | 30 km/h     | Medium     |
| 11 | clinic   | Dr. Saravanakumar Clinic            | 11.5842002, 78.4441223 | 30 km/h     | Medium     |

### Crowded Spots

|   | type        | name                                      | coordinates            | speed_limit | risk_level |
|---|-------------|-------------------------------------------|------------------------|-------------|------------|
| 2 | school      | அரசு ஆண்கள் மேல்நிலைப் பள்ளி              | 11.3850439, 77.8948279 | 30 km/h     | Medium     |
| 3 | school      | அரசு பெண்கள் மேல்நிலைப் பள்ளி             | 11.3844247, 77.8949053 | 30 km/h     | Medium     |
| 4 | marketplace | திருச்செங்கோடு தினசரி காய்கறி சந்தை       | 11.3833608, 77.8970145 | 30 km/h     | Medium     |
| 6 | marketplace | Farmer Market                             | 11.4146849, 78.0805463 | 30 km/h     | Medium     |
| 7 | school      | Government Higher Secondary School        | 11.4150045, 78.0838172 | 30 km/h     | Medium     |
| 8 | school      | Government school                         | 11.4248027, 78.1323902 | 30 km/h     | Medium     |
| 9 | college     | Paavai Arts And Science College For Women | 11.4262443, 78.1546987 | 30 km/h     | Medium     |



## Route Photos of Risky Spots



**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.38395, 77.89480



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44019, 77.87547





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.39685, 77.88861



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.38346, 77.89975





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.38302, 77.89995**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.38188, 77.90257**



**Risk Type: Blind Spot**

**Risk Level: Blind Spot**

**Speed Limit: 10 KM/Hr**

**Coordinates: 11.37841, 77.90140**





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.36745, 77.93951



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.41361, 78.08073





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.41453, 78.08062



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.41476, 78.08094



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.42956, 78.11590





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.42450, 78.15278



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.43542, 78.20754





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.45075, 78.20391**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.45099, 78.20397**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.44946, 78.22270**





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44793, 78.22320



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.44753, 78.22443





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45361, 78.23597



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.49628, 78.32342



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.50514, 78.33092





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.50868, 78.33972



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.51207, 78.34431





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.51872, 78.34817



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.56722, 78.37819



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.56945, 78.39881





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.58331, 78.44344



**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.58139, 78.46967





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.61424, 78.47467

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